

Training programming and prescription

with DARREN BURGESS

In football as in watchmaking, talent and elegance mean nothing without rigour and precision.

Lionel Messi

The sports medicine clinician should understand the different elements of training and their possible relationship to injury. This facilitates the practitioner's obtaining a full training history from an injured athlete, learning about training strategy from a coach or fitness practitioner and enhances a clinician's understanding of the phases of rehabilitation outlined in Chapter 18. This chapter reviews the principles of training and outlines some more common training programming and assessment practices. The reader is directed to other sources for more detailed outlines of the various types of training.

PRINCIPLES OF TRAINING

'Training' is the pursuit of activity that will ultimately lead to an improved performance in a given sport. A number of general principles of training apply to all sports:

- periodisation
- overload
- specificity
- individuality.

Periodisation

Periodisation is an important component of all training programs, in both the long and short term. Training can be divided into three distinct phases: conditioning (preparation), pre-competition (transitional) and competition.

Generally, the conditioning phase emphasises developing aerobic and anaerobic fitness, strength and power. Often during this period, the athlete is training under fatigue and if required to compete would probably perform poorly. During the pre-competition phase of training, the emphasis switches from pure conditioning to more technical work. During the competition phase, the emphasis is on competitive performance while maintaining basic conditioning (Table 10.1).

In many sports (e.g. football, basketball, hockey), a 4–6-month competition season is usual. In some instances, an athlete is required to undertake two periods of competition in the one year. A suggested program for athletes in these two situations is outlined in Figure 10.1. In other instances, the competition period may last as long as 8–10 months and conditioning work can extend into the competitive season. However, in all of these scenarios the same principles of training periodisation apply.

To ensure complete recovery from the physical and mental stress of competition, adequate time should be allowed between the end of one season or competition phase and the start of the next season or phase. This period may last 4–6 weeks but is dependent on when the next competition begins.

In the intermediate time frame, it is important to introduce easy weeks into the training program; these give the athlete time to recover (Chapter 13) and diminish

Table 10.1 Different types of training are performed during the three phases of the yearly cycle

Training phase	Aerobic training	Anaerobic training	Plyometrics training	Weight training	Technique training
Preparation/conditioning	+++	++	++	+++	+
Transitional/pre-competition	++	+++	++	++	+++
Competition	+	+	–	+	++

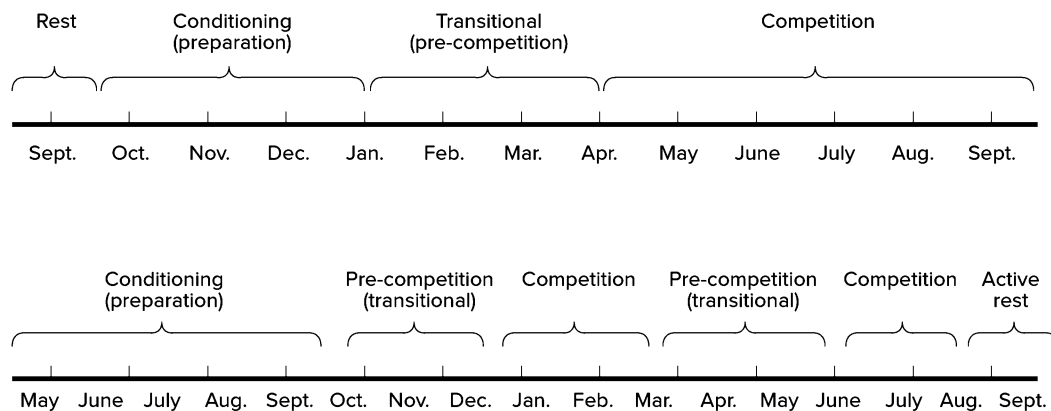


Figure 10.1 Periodisation of training showing a single cycle annual program (above) and a dual cycle annual program (below)

risk of injury. During these easy weeks, the volume of training is typically reduced; however, intensity should be maintained to prevent detraining.¹

In the short term, the training program must allow for adequate recovery between training sessions. This ensures the athlete is able to train at appropriate intensities throughout the week and reduces the risk of injury.¹

Overload

Overload is a variable that coaches manipulate to allow the athlete to perform work at a greater intensity or to perform a greater volume of work at a given intensity. Practically speaking, overloading an athlete involves applying stress to the body over and above that which is normally, or has been recently, encountered. If increased stress is not excessive and adequate adaptation time is allowed, the work capacity of the athlete will be increased ('supercompensation'). Athletes should be carefully monitored during periods of overload to prevent injury, overtraining and even poor performance from occurring.

Specificity

Specificity refers to the principle of directing training to performance in the athlete's given sport. It is important, therefore, to identify the most important components of fitness for each particular sport and to tailor the athlete's

training toward improving these particular components. For example, there is no advantage for a strictly power athlete in doing large amounts of endurance training.

Specificity can refer to both training the specific fitness demands of a sport as well as training the direct movement patterns of the sport to improve the fitness of the athlete. Those choosing the latter method argue that specific training in this way has the advantage of training skills and decision making while improving fitness.² Most field sports (e.g. football) require a complex combination of both strength and endurance training and therefore specificity should include a combination of specific fitness training as well as training the specific movements of a sport.

Individualisation

As individual differences between athletes are great, training must be tailored to the individual's needs. Individuals differ in their tolerance of particular training loads, response to specific training stimuli, speed of recovery, psychological make-up, nutritional intake and lifestyle habits. Individual responses to training are influenced by previous training history, age, current state of fitness, genetic make-up and so on. Even in a team setting, it is vital that players are treated as individuals and trained accordingly.

The Central Governor Model (CGM) for the limits of performance and its relevance to interval training

The concept of maximum oxygen consumption (VO_2max) and lactic acidosis limiting athletic performance has undergone much critical evaluation. The 'classical' model of Hill,³ which still enjoys support among a substantial number of exercise physiologists, suggests that:

- Progressive muscle hypoxia limits maximal exercise performance. As a result, the main determinant of exercise performance is the heart's ability to supply sufficient blood (and oxygen) to the exercising muscles.

- Anaerobiosis (lack of muscle oxygen) secondary to the inability further to increase the cardiac output (producing a 'plateau' in cardiac output) explains the onset of lactate production by skeletal muscle at the 'anaerobic threshold'.
- Mitochondrial adaptation in the exercising muscles, associated with an increased ability of the heart to pump a larger cardiac output, are the exclusive biological changes that explain changes in performance with endurance training.

This model has been challenged by the contemporary, but certainly not universally accepted, model of Professor Timothy Noakes. Noakes' data refute the classical model. Instead, he proposes that skeletal muscle recruitment and contractile function are regulated by a hierarchy of controls (conceptually 'the central governor') specifically to prevent damage to any number of different organs.⁴⁻¹⁴ He, and subsequently others, argues that according to the Hill model in which a plateau in cardiac output precedes the development of skeletal muscle anaerobiosis and lactic acidosis, the first organ to be threatened by the plateau in cardiac output would be the heart, not skeletal muscle. The plateau in cardiac output would prevent any further increase in blood flow to the heart, leading

to myocardial ischaemia, the onset of chest pain (angina pectoris) and heart failure.

He also provides evidence that this was, in fact, the belief of the early exercise physiologists, including Hill in England and Dill at the Harvard Fatigue Laboratory in the United States and was a central component of their teachings. Since Hill understood that the heart could not survive a prolonged period of ischaemia, he conceived the presence of a 'governor' in either the heart or the brain that would reduce heart function and spare the heart by causing a 'slowing of the circulation' as soon as myocardial ischaemia developed. Noakes and colleagues have extended this interpretation to suggest that the governor exists in the central nervous system, hence it is called the central governor and that it responds to multiple sensory inputs from all the organs in the body (Fig. 10.2). In response to those inputs, the central governor regulates the number of motor units that can be recruited in the exercising limbs on a moment-to-moment basis, reducing or limiting the number that can be recruited when their continued recruitment, necessary to maintain the work output or exercise intensity, threatens whole-body homeostasis.

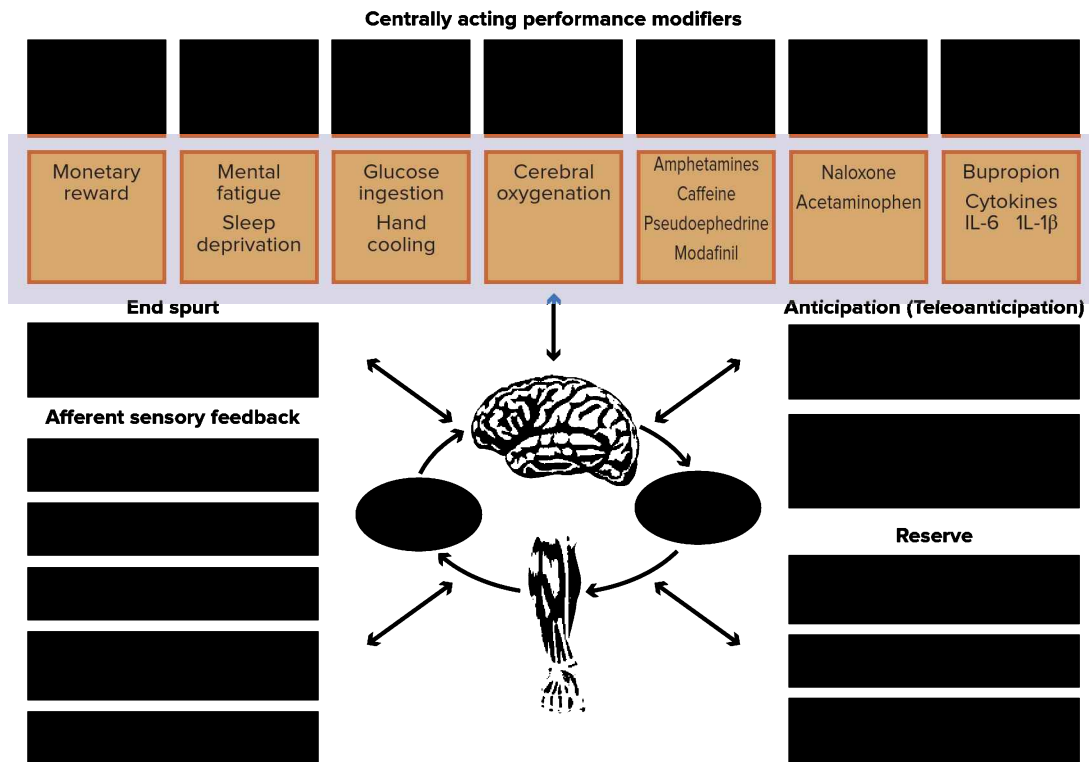


Figure 10.2 The central governor responds to multiple sensory inputs from all organs of the body

continued

Hence, Noakes writes that 'during maximal exercise, progressive myocardial ischaemia preceding skeletal muscle anaerobiosis must be thwarted, so that neither the heart nor the skeletal muscle develop irreversible rigour and necrosis with fatal consequences'. The reader is directed to the publications that summarise this argument to date.^{4, 7–14}

Relevance to interval training

Irrespective of the theoretical background that underpins the physiology, the most efficient method of increasing anaerobic fitness is a form of intermittent exercise or interval training. Interval training involves a number of bouts of exercise separated by periods of rest or recovery.

The principle of such training is to achieve a level of lactic acidosis with one individual effort and then allow the body to recover from its effects before embarking on another bout of

exercise. There is scope for enormous variation in the intensity and duration of the exercise bouts and the duration of the recovery period. Anaerobic training will also reset the central governor mechanism, perhaps teaching the governor that it can allow a slightly higher exercise intensity without risking a catastrophic bodily failure. In addition, interval training will also increase the ability of the skeletal muscle fibres to produce more force. There is some evidence suggesting that differences in skeletal muscle contractility may modify athletic performance by determining the shortest possible duration the foot is in contact with the ground. This is because faster running requires shorter foot contact times and hence greater skeletal muscle contractility and the recruitment of a greater number of muscle fibres.¹⁵ In contrast, skeletal muscle contractility may be impaired in certain disease states.¹⁶

CONDITIONING TRAINING

The maintenance of fitness is not only crucial to sporting success and injury prevention it is also an essential component of the rehabilitation process. Regardless of the injury the athlete has sustained, exercises to maintain fitness should be incorporated as soon as possible. For example, with injuries to the lower limb, cardiovascular fitness may be maintained by performing activities such as swimming with a pool buoy or arm 'grinder' work. Depending on the athlete's particular sport, this may include a combination of endurance, interval, anaerobic and power work. It is important to maintain alternative training methods for cardiovascular fitness, to encourage motivation and compliance with general fitness goals.

We have divided conditioning into five separate types of training—endurance, speed, agility, resistance and flexibility—and will discuss each of these separately. In practice these methods are often combined for a more global conditioning effect.

Endurance training

Technically, endurance training aims to increase oxygen delivery to the working muscles and thereby increase aerobic capacity. Adaptations to endurance training include an increase in maximal oxygen uptake (VO_2max) and an increased ability of skeletal muscle to generate energy via oxidative metabolism.^{17, 18} (For a more detailed discussion of the debate on VO_2max and the role of lactic acidosis in limiting training, see box below).

The long-term consequences of prolonged endurance training have been debated in recent years, with proposed cardiovascular risks associated with cardiac fatigue.^{19, 20} However, a study of young Olympic athletes subjected to extreme uninterrupted endurance training over prolonged periods (up to 17 years) found that endurance training was

not associated with significant changes in left ventricular morphology, deterioration in left ventricular function, or occurrence of cardiovascular symptoms or events.²¹ Endurance sports include long-distance running (greater than 5 km), cross-country skiing, cycling, rowing and triathlon.

It is important here to be aware of the many benefits an increased aerobic capacity can have to a range of different sports. An enhanced aerobic capacity will not only improve the efficiency of oxygen delivery but will also improve the regeneration of anaerobic energy pathways. This means performance in intermittent sports such as ice hockey, soccer, American football and netball, all traditionally thought of as strictly anaerobic sports, will be improved with an enhanced aerobic capacity. This is important to remember when prescribing or reviewing rehabilitation programs for these sports.

Endurance training methods

There are numerous methods to train endurance capacity. This section focuses on the more popular techniques.

Interval training

Interval training involves numerous bouts of high intensity efforts of set times or distances. Both effort duration and recovery time in between efforts dictate the intensity of each bout. An endurance interval session might be 6 sets of 1 minute running efforts. For a high-intensity session, 2 minutes recovery could be prescribed in between efforts. To challenge the aerobic capacity to recover more quickly, 1 minute recovery between sets could be prescribed. These repeat-effort intervals have been shown to dramatically increase VO_2max .²²

Fartlek training

'Fartlek', which means 'speed play' in Swedish, is a popular form of endurance training that involves high-intensity efforts

interspersed between constant paced exercise. A typical fartlek session on a bike might involve 30 seconds of high-intensity cycling every 2.5 minutes for 30 minutes. This example would provide the cyclist with 10 high-intensity efforts during a 30-minute ride. In some sports, athletes may be unaware of the duration and intensity of efforts and have to respond to the coach's feedback during the session.

Maximal aerobic speed (MAS) training

MAS training involves calculating an athlete's maximal aerobic speed and then prescribing a certain percentage of this within an interval training session.²³ Typically, the speed an athlete can complete a 2 km (or similar distance) time trial is used as that athlete's MAS. Precise distances and expected times can then be calculated for each athlete. The most common time periods for MAS are 15 seconds of work followed by 15 seconds of exercise (15:15).

Cross-training

To prevent injury, it may be beneficial to reduce the amount of weight-bearing exercise. Cross-training enables the athlete to maintain aerobic fitness while reducing stress on weight-bearing joints, muscles and tendons. In athletes with a chronic condition such as articular cartilage damage to a weight-bearing joint, cross-training may be used to reduce the impact load while maintaining adequate training volume. Similarly, in a patient returning to sport from an overuse injury (such as a stress fracture), cross-training can reduce the risk of recurrence. Runners may wish to introduce one to two sessions per week of activities such as cycling, swimming or water running. These alternative work-outs can mirror the athlete's usual training session (e.g. interval, fartlek).

Skill training as endurance

It is important to realise that a skill drill or indeed entire skill session can enhance endurance capacity. In particular, there are numerous ways to alter skill drills to emphasise endurance, while maintaining the important aspects of the skill (e.g. decision making, team combinations, specific skill execution). For example, a simple soccer game could be altered in any of the following ways to increase the aerobic requirements of the exercise:

- Increase the size of the field.
- Decrease the playing numbers.
- For a goal to be scored all players must be over the half-way line.
- Every 1 minute the coach blows a whistle and one player from each team must run to a specified marker outside of the field.

Speed training

Running speed, which is an important component of many sports, is a largely inherited ability. However, athletes can

improve their speed by enhancing muscular power and strength, as well as by improving technique, which increases the efficiency of ground coverage. Therefore, running speed can be increased by undertaking resistance and power training as well as by performing technical running drills.

Speed training methods

There is much debate about how much speed can be developed or improved in the mature athlete. Speed can arguably be enhanced, or at the very least maintained, with an appropriate strength and power program—this will be discussed later in this chapter. However, debate remains about the extent to which speed can be improved through either technical modification or repeated exposure to a speed stimulus. Both of these will be discussed briefly below.

Technique training

Improving the technical aspect of speed requires significant time spent analysing athlete running mechanics and then performing extensive corrective exercises as required. Within most sports, time spent dedicated specifically to technical speed work might often be at the expense of other conditioning, strength training or even skill training. As a result, most speed technique work is often confined to small amounts of time with large groups of players and/or warm-up activities pre-training/strength sessions. This requires a fairly broad approach where emphasis is placed on high knee drive, exaggerated heel lift and horizontal foot position throughout the sprint motion. An ideal opportunity exists to improve speed technique during the rehabilitation process where a more individualised approach can be taken. Of course, in sports such as running and long jump, a large percentage of training time should be devoted to technical speed drills.

Speed intervals

Appropriate exposure to speed will not only maintain speed qualities but may also prevent soft tissue injury,²⁴ particularly in the hamstrings. These 'unpractised' movements that occur in games often cause muscle injury.

PRACTICE PEARL

Regular speed interval work, particularly during weekly competition cycles will ensure the muscles and tendons are not 'shocked' into unfamiliar explosive movements that may occur during games.

Practitioners should remember that absolute speed intervals should be performed while the athlete is fresh; however, speed exposure in training should also occur under fatigue to prepare the body to what it may encounter during competition.

Skill training as speed

In most sports, speed training can and should be integrated into skill training. This can be easily achieved by using execution of simple skills at the end of sprinting activities. Care should be taken in programming these drills as they can be quite physically and structurally demanding.

Agility training

Agility can be defined as the ability to alter direction to achieve a specific goal (e.g. evade/deceive/react to an opponent, create space). Agility and rapid reflexes are often inherited characteristics. However, like speed, they can be improved somewhat by training and, thus, are included in training programs of all sports. Agility training has also been implemented in seniors to prevent falls.²⁵

Agility training methods

Agility is often incorrectly trained using stationary objects (e.g. cones, poles, ladders). This may allow the coach to teach the athlete the correct foot placement or stride mechanics; however, agility training requires a reaction and therefore a decision-making component.²⁶

Technique training

Training agility technique may be appropriate in the initial stages of rehabilitation or perhaps within adolescent programs. The coach can spend some time on effective techniques to approach a known object. However, this type of training should be limited as it often leads to inappropriate habits such as a player keeping their head down to review foot placement.

Agility drills

A reaction component should always be included in agility drills. This can be reacting to a coach movement, coach command, opponent or teammate. Advanced agility drills should also include a decision-making element.

Skill training as agility

Most sports skill training could be counted as agility training. This is because the majority of field sports training involves reacting to opponents, the ball or teammates. This means any additional agility-specific training should be programmed carefully as repeated changes of direction could cause some overload issues to the pelvic region.

Resistance training

Resistance training can be used to enhance athletic performance, improve musculoskeletal health and correct muscle imbalances.²⁷ Resistance training is often used in rehabilitation when weakness compromises function and sports performance. This is particularly true

following periods of immobilisation or injury and in pain presentations.

The primary goals of resistance training are normally to increase muscle size (hypertrophy), strength or power. Increases in muscle hypertrophy and strength are dependent on five biochemical and physiological factors that are all stimulated and enhanced by appropriate resistance training:

- increased glycogen and protein storage in muscle
- increased vascularisation
- biochemical changes affecting the enzymes of energy metabolism
- increased number of myofibrils
- recruitment of neighbouring motor units.

It is important that the athlete is provided with the correct environment in which to perform resistance training. The following factors will help maximise gains during resistance training:

- adequate warm-up to increase body temperature and metabolic efficiency
- good quality, controlled performance of the exercise
- pain-free performance of exercise
- use of slow, controlled exercise initially, with little or no resistance, to develop a good base for neural patterning to occur
- comprehensive stretching program to restore/maintain full range of motion
- muscle strengthening throughout the entire available range of motion.

Additionally, there is a training crossover effect—when one limb is trained, strength gains will also be recorded in the contralateral limb.^{27, 28} This phenomenon is termed 'central adaptation' and reflects the important role the central nervous system plays in motor unit firing, as well as the bracing role the contralateral limb plays in most single-limb exercises.

Types of resistance exercises

The three main types of exercise used in resistance training are:

- isometric
- isotonic
- isokinetic.

Isometric exercise

An isometric exercise occurs when a muscle contracts without associated movement of the joint on which the muscle acts. Isometric exercises are often the first form of strengthening exercise used after injury, especially if the region is excessively painful or if the area is immobilised. These exercises are commenced as soon as the athlete can perform them without pain.



Figure 10.3 Isometric co-contraction of the hamstrings, gluteal and quadriceps muscles with the patient pushing foot into wall

Isometric exercises are used when a muscle is too weak to perform range of motion exercises, in conditions where other forms of exercise are not possible (such as patellar dislocation and shoulder dislocation) or when isometric contraction is required in activities (e.g. stabilising). Isometric exercises can minimise muscle atrophy associated with immobilisation and injury by maintaining or improving static strength, minimising swelling via the muscle pump action, and enhancing neural and proprioceptive inputs to the muscle.

Ideally, isometric exercises are initially held for 5 seconds with contraction time increasing as the athlete becomes more tolerant. They can be performed frequently during the day as pain permits. The number of sets will vary at different stages of the rehabilitation program. If an athlete has difficulty, exercises may be performed against resistance or against an immovable object. It is important to remember that the quality of exercise is more important than the quantity.

Isometric exercises should be carried out at multiple angles if possible, as strength gain can be specific to the angle of exercise. The athlete should progress from submaximal to maximal isometric exercise slowly within the limitations of pain and function.

Isometric exercises have demonstrated success in the rehabilitation of tendon injuries (see Chapter 4) and should form a large component of any musculotendinous rehabilitation program.²⁹

When significant isometric effort is tolerated at multiple joint angles, dynamic exercises may begin. An example of an isometric exercise for the lower limb is shown in Figure 10.3.

Isotonic exercise

Isotonic exercises are performed when the joint moves through a range of motion against a constant resistance

or weight. Isotonic exercises may be performed with free weights, such as dumbbells, pulleys or sandbags (Fig. 10.4).

Free weights encourage natural movement patterns and require muscle coordination and joint stability in all planes of movement and therefore may transfer strength gains more readily to the playing field.³⁰ With free weights it is possible to simulate athletic activities as the body position can be varied. Isotonic exercises may be:

- concentric—a shortening isotonic contraction in which the origin and insertion of the muscles approximate. Individual muscle fibres shorten during concentric contraction



(a)



(b)

Figure 10.4 Isotonic exercises (a) Dumbbell (b) Sandbag



Figure 10.5 Concentric (white arrow) and eccentric (black arrow) quadriceps exercises

- eccentric—a lengthening isotonic contraction where the origin and insertion of the muscles separate. The individual muscle fibres lengthen during eccentric contraction.

Concentric and eccentric exercises for the quadriceps are shown in Figure 10.5.

The intramuscular force produced per motor unit during an eccentric contraction is larger than that during a concentric contraction.²⁸ Eccentric contractions may generate high tension within the series elastic component, which consists of connective tissue and the actin-myosin cross-bridges in muscles. It has been observed that eccentric exercise results in higher rates of delayed onset muscle soreness (DOMS) and even muscle damage if used inappropriately.^{31, 32} Consequently, eccentric programs should commence at very low levels and progress gradually to higher intensity and volume. The use of eccentric exercise programs may help prevent recurrence of musculotendinous injuries.

Eccentric training has been advocated in the rehabilitation of tendon injuries, due to the proposed facilitation of tendon remodelling through promotion of collagen fibres within the injured tendon.³²⁻³⁴ There is evidence that Achilles, patellar and lateral elbow tendinopathies respond well to an eccentric rehabilitation program.^{31, 32, 34, 35}

However, not all tendon injuries benefit from eccentric exercises. Specifically, eccentric exercises have shown poor success rates for rotator cuff tendinopathy³⁶ and for insertional Achilles tendinopathy, compared to mid-tendon Achilles lesions.³² Therefore, the site of tendon pathology should be considered when prescribing eccentric exercises.

Practically, there are some potential dangers of isotonic resistance training such as:

- Athletes require adequate supervision in the gymnasium.
- Athletes should never attempt to lift a maximal weight without a 'spotter' (an assistant who is able to help the athlete if problems arise).
- Isotonic machines such as Keiser[®] equipment may provide a safe alternative to free weights, but these machines limit the range of motion and are generally unable to provide truly constant resistance through the lift.

Isotonic exercises in which the body weight of the individual is used as resistance are also safer than free weights and are often more convenient to perform. Exercises such as sit-ups, push-ups and chin-ups can be done almost anywhere and require no supervision. However, it is difficult to increase the resistance of the exercise and the only way to increase the effort is to increase the number of repetitions performed.

Isokinetic exercises

Isokinetic exercises are performed on devices at a fixed speed with a variable resistance that is totally accommodative to the individual throughout the range of motion. The velocity is, therefore, constant at a preselected dynamic rate, while the resistance varies to match the force applied at every point in the range of motion. This enables the patient to perform more work than is possible with either constant or variable resistance isotonic exercise.

Isokinetic testing can highlight imbalances, such as scapular muscle imbalances in overhead athletes with chronic impingement signs.³⁷ A number of isokinetic devices are available and include the Ariel[®], Biodex[®], Cybex[®], KinCom[®], Lido and Merac[®] machines. However, these machines do come at a cost, which may explain why they are more commonplace in research than in clinical settings.

Classification of resistance exercises

Open chain and closed chain exercises

An open kinetic chain exercise often involves single joint movement performed in a non weight-bearing position where the distal extremity freely moves through space.³⁸ Closed kinetic chain exercises involve multiple joints and

Table 10.2 Advantages and disadvantages of open chain and closed chain exercises

	Advantages	Disadvantages
Open chain exercises	Decreased joint compression Can exercise in non weight-bearing positions Able to exercise through increased ROM Able to isolate individual muscles	Increased joint translation Decreased functionality
Closed chain exercises	Decreased joint forces in secondary joints (e.g. less patellofemoral force with squat) Decreased joint translation Increased functionality	Increased joint compression Not able to exercise through increased ROM Not able to isolate individual muscles

ROM = range of motion

are performed in weight-bearing positions with a fixed distal extremity. Closed kinetic exercises are thought to be more functional, provide more proprioceptive feedback and cause less shear joint forces than open kinetic chain exercises.³⁸

Open and closed chain exercises in rehabilitation
Although some studies promote closed chain over open chain exercises,^{38, 39} others advocate that both types of exercises play beneficial roles in rehabilitation, especially in regards to anterior cruciate rehabilitation and patellofemoral pain.^{40, 41, 42} Proposed advantages and disadvantages of open and closed kinetic exercise are shown in Table 10.2.

Examples of these exercises are shown in Figure 10.6. Figure 10.6c shows an example of open (right arm) and closed (left arm) chain exercises for the shoulder girdle.



(a)

Figure 10.6 Open and closed chain exercises (a) Open chain knee extension with the foot moving freely



(b)



(c)

Figure 10.6 (cont.) (b) Closed chain knee extension with the feet immobile (c) Open chain (right arm) and closed chain (left arm) exercises on an unstable surface

Closed chain upper limb exercises are particularly useful during the early recovery period from shoulder surgery (Chapter 24). Excessive mobility and compromised static stability within the glenohumeral joint have been linked to capsular, labral and musculotendinous injuries in throwing athletes.⁴³ The positive benefits of closed kinetic chain exercises, performed under load-bearing positions, are thought to stimulate joint receptors and facilitate muscle co-contractions around the shoulder and therefore enhance joint stability.^{37, 43}

Qualities of resistance training

This section will briefly describe the various resistance training qualities and outline the methods the clinician or coach may employ to train these qualities.

Strength

Muscle strength is the muscle's ability to exert force. Strength gains can be seen quickly, even before physiological hypertrophy occurs. The initial strength gain in response to exercise is thought to be related to increased neuromuscular facilitation (i.e. the nervous system enhances the motor pathways so that the muscle group becomes more neurologically efficient).^{27, 28, 44} Neural adaptations facilitate changes in coordination and learning that enhance the recruitment and activation of muscles during a strength task.²⁷

Typical loading patterns for strength training can be seen in Table 10.3. Most importantly the resistance needs to be at or near maximal levels with large rest periods in between sets. This enables the muscles to recuperate adequately in between sets.

Strength training is generally the base from which all other resistance training emanates and therefore it generally occurs at the beginning of pre-season training.

Power

Muscle power is the muscle's rate of doing work. Similar to strength training, the initial power gains observed with power training result from improvements in neuromuscular efficiency.⁴⁵ Specifically, initial improvements in power can be attributed to improved muscle coordination between agonist and antagonist muscles. Power exercises may include:

- fast-speed isotonic or isokinetic exercises (concentric and/or eccentric)
- increased speed of functional exercises (e.g. faster reverse calf raise, drop squat)
- plyometric exercises (e.g. hopping, bounding).

Power exercises often involve functional and sport-specific exercises. Exercises should be made appropriate to the athlete's sport to gain optimal benefits (e.g. bounding for a sprinter, jump and land for a basketballer).

Another technique of increasing power is plyometric training. Plyometric exercises (plyometrics) use the natural elastic recoil elements of human muscle and the neurological stretch reflex to produce a stronger, faster muscle response. Plyometrics combine a rapid eccentric muscle contraction with a rapid concentric contraction to produce a fast forceful movement. It must be performed in conjunction with a resistance training program, as athletes need to have minimum basic strength levels before commencing plyometrics.

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Because of their explosive nature, plyometrics have a great potential for injury. Therefore, an athlete's plyometrics program should be carefully supervised.

Plyometric exercises include hopping and bounding drills, jumps over hurdles and depth jumps. These activities emphasise spending as little time as possible in contact with the ground. This form of exercise can cause DOMS (Chapter 4). Plyometric training should only be performed when the athlete is fresh and the volume of work should be built up gradually. The training surface must be firm but forgiving, such as sprung basketball floors. When technique begins to deteriorate, the exercise should be stopped.

Olympic-type weightlifting is often used as part of a power training program. Olympic lifting involves explosively lifting a weight from the floor to a position above the ground using the entire body. Typical Olympic lifts include the power clean, snatch, and clean and jerk. These lifts exercise a greater number of muscle groups

Table 10.3 Resistance training loading patterns

	Sets	Reps	Rest (s)	%RM	Speed
Strength	3–5	3–6	90–120	90–100	Controlled
Power	3–4	8–12	45–60	40–60	Explosive
Hypertrophy	2–3	8–12	60–75	70–90	Slow on eccentric
Endurance	1–2	15–30	45–60	50–70	Sport-specific

than conventional weightlifting, exercising them both concentrically and eccentrically. The potential for injury is high and athletes must learn correct lifting techniques before attempting these lifts.

When injury has decreased muscle power or the athlete's sport includes periods of explosive power, the rehabilitation program should incorporate power exercises. Commonly, power-focused exercises are incorporated into the later stage of rehabilitation due to the potential for re-injury and explosive nature of this training.

Typical loading patterns for a power program can be seen in Table 10.3. The most important factor in a power development program is the speed of movement which must be as explosive as possible.

Muscular endurance

Muscle endurance is the muscle's ability to sustain contraction or perform repeated contractions. The aim of muscular endurance training is to increase the capacity to sustain repetitive, high-intensity, low-resistance exercise such as running, cycling and swimming.¹⁷ Excessive bouts of muscular endurance training can result in fatigue and reduced sporting performance. Therefore, care should be taken in incorporating this type of training into a resistance training program, especially during prolonged pre-season training.

Typical loading patterns for muscular endurance training can be seen in Table 10.3. This type of training involves high repetitions with very little rest.

Hypertrophy training

Hypertrophy training aims to increase the muscle size. This quality is important particularly for combative sports where an increased muscle mass can be advantageous. Generally, hypertrophy occurs through an increase in the size or cross-sectional area of the muscle cells rather than an increase in the number of muscle cells.

Typical loading patterns for hypertrophy can be found in Table 10.3. As this table demonstrates, the eccentric or lowering component should be slower than the concentric component. This will create additional muscle damage and therefore take greater time for the muscle to repair and regenerate. As a result care must be taken when prescribing additional speed or resistance training around hypertrophy sessions.

Flexibility training

An athlete is susceptible to injury if they lack sufficient flexibility to meet their sport-specific demands. Stretching and range of motion exercises are performed to increase joint range of motion. The acute effects of stretching include:^{46, 47}

- decreased muscle stiffness through viscoelastic deformation of the muscle tissue (this is likely to be extremely transient)
- increased muscle length through serial addition of sarcomeres
- altering sensation, thereby increasing stretch tolerance.

Traditionally, stretching has been widely promoted for injury prevention and performance enhancing for sporting activity. However, some authors have suggested stretching does not prevent injury in otherwise healthy individuals.⁴⁸ Others have noted that it is important to differentiate between pre-exercise stretching (where stretching does not appear to prevent injury)⁴⁹ and regular stretching outside periods of exercise (where there is some clinical and basic science evidence suggesting stretching may prevent injury).^{50, 51} Additionally, stretching does not seem to reduce the effects of DOMS.⁵²

Furthermore, the acute effects of stretching can cause temporary strength deficits and hence reduce sporting performance.⁵³ This deficit seems to be associated with stretches performed for more than 60 seconds, while stretches of shorter duration may have less significant deficits.⁵³ As traditional stretching routines are performed during warm-up sessions prior to playing sport, the amount of time athletes stretch should be taken into consideration.

Stretching is often promoted as having a number of beneficial effects,^{54, 55} although not all of the following have been appropriately studied:

- increases muscle and joint flexibility
- increases muscle relaxation
- decreases muscle soreness
- improves circulation
- helps prevent excessive adhesion
- promotes a flexible, strong scar.

Types of stretching

There are three main types of stretching: static, dynamic and proprioceptive neuromuscular facilitation.

Static stretching

Static stretching involves moving the muscle or joint into an elongated position and holding the position for an extended period. Historically, this type of stretching has been used to prepare the muscle for exercise; however, research has consistently demonstrated a reduction in power immediately after static stretching.⁵⁶ As a result the more contemporary practice for athletic populations is to use static stretching both post training for recovery and away from training to improve flexibility.

Recommendations for effective stretching

- A gentle warm-up before stretching increases tissue temperature and facilitates stretching. This may include activities such as jogging or cycling.
- As with warm-up, superficial or deep heat modalities may be applied to the area prior to stretching to increase tissue temperature.
- Cryotherapy may reduce pain and muscle spasm and thereby enhance the overall stretch of a muscle in the initial stages after an injury, even though the temperature is decreased (i.e. opposite to heat).
- Athletes should be carefully instructed regarding the correct stretching position, as incorrect positioning may cause injury.
- Different muscles seem to require different durations of stretch. In general, a slow sustained stretch should be held for a minimum of 10–15 seconds and progressed for 1 minute or longer. The athlete should feel stretch in the appropriate area.
- As the athlete's flexibility improves, increases in intensity, duration, frequency and type of stretch can be considered.
- Stretching should always be pain-free, that is, 'tightness' without pain.

Dynamic stretching

Dynamic or ballistic stretching is when the muscles and joints are taken through their range of motion during movement. This type of practice is more specific to preparation for exercise and sports in particular. During the rehabilitation process care should be taken to not 'bounce' the muscle that is recovering from injury. Dynamic stretches have been shown to significantly increase tendon flexibility and elasticity and have been promoted for end-stage rehabilitation for tendon injuries. However, ballistic stretching involves eccentric contractions during the stretch phase, which may result in soreness or injury and therefore care should be taken when incorporating such stretches.

Proprioceptive neuromuscular facilitation (PNF) stretching

PNF stretching involves the combination of contracting the antagonist (opposite) muscle and excessively stretching the agonist muscle. This type of stretching is quite aggressive but can result in rapid increases in flexibility. PNF flexibility sessions should not be performed more than 1–2 times per week and should be cycled into and out of training as it can cause some muscle damage.

Flexibility in the rehabilitation process

Regaining or maintaining full flexibility of joints and soft tissues is an essential component of the rehabilitation process. Following injury, musculotendinous flexibility often decreases as a result of spasm of surrounding muscles. Inflammation, pain and/or stiffness can limit joint range of motion and the normal extensibility of the musculotendinous unit can be compromised. This may result in dysfunction of adjacent joints and soft tissues.

For example, the lumbar spine may be restricted in range of motion and the paraspinal muscles may spasm following knee or hip surgery, especially after periods of restricted mobilisation. Adequate soft tissue extensibility after injury is essential to encourage pain-free range of movements. Stretching muscles and joints is one way of improving tissue and joint extensibility.

TRAINING LOAD MANAGEMENT FOR PERFORMANCE ENHANCEMENT

The ability to effectively periodise and manage training load will obviously have a large impact on athlete success. Long-term, gradually progressive training plans should be made, including periods of higher and lower loading to avoid improvement plateaus, injuries, illnesses or burn-out. This process, commonly referred to as *periodisation*, should consider time periods consisting of several months (macrocycles), weeks (mesocycles) and days (microcycles).

Load management not only comprises sound knowledge of training periodisation, but also an intimate awareness of when players are fatigued, undertrained, stressed or fresh. Typically, team-sport athletes have been advised to avoid strenuous mid-week training to 'freshen up' for the next game and/or avoid injury. However, recent evidence suggests that too little training can be a significant risk factor for injury.⁵⁷ This suggests a delicate balance exists between fitness, fatigue and freshness, with the ability to identify each state being crucial to performance enhancement and injury prevention.

To successfully manage load, it needs to be quantified. There are, of course, many influences on the total load placed upon an athlete, and the ability to identify these and alter training load accordingly is both an art and a science. Certainly, technology may assist the practitioner in quantifying load; however, just as important is the

ability to 'sense' or 'feel' when players are fatigued or fresh. This 'sense', however, typically requires intimate knowledge of the players/athletes and much experience in this domain.

Training load is typically defined as either internal or external. Internal load refers to the impact the training has on the athlete and can be measured by subjective wellness questionnaires,^{58, 59} as well as a range of objective metrics such as heart rate, blood markers (e.g. creatine kinase, testosterone, cortisol), RPE and heart rate variability.⁵⁹ External load refers to the estimation of the actual load the players are placed under and is usually measured using GPS devices, power meters or simply calculating training duration.⁵⁹

Maximising training output (external load) while minimising training cost (internal load) is usually the goal of most training programs. Effectively estimating both of these, therefore, becomes a crucial element of the sports practitioner's role. RPE-load (also called session-RPE) has subsequently gained much popularity in both research⁶⁰ and practice as it represents both internal and external load and allows different forms of training (e.g. football and weightlifting) to be measured in the same units. RPE-load is calculated by multiplying player RPE (internal) with training or match duration (external) and has been shown to relate effectively to measures of internal load.⁶¹

The role of load monitoring in injury prevention is covered in Chapter 12.

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e

